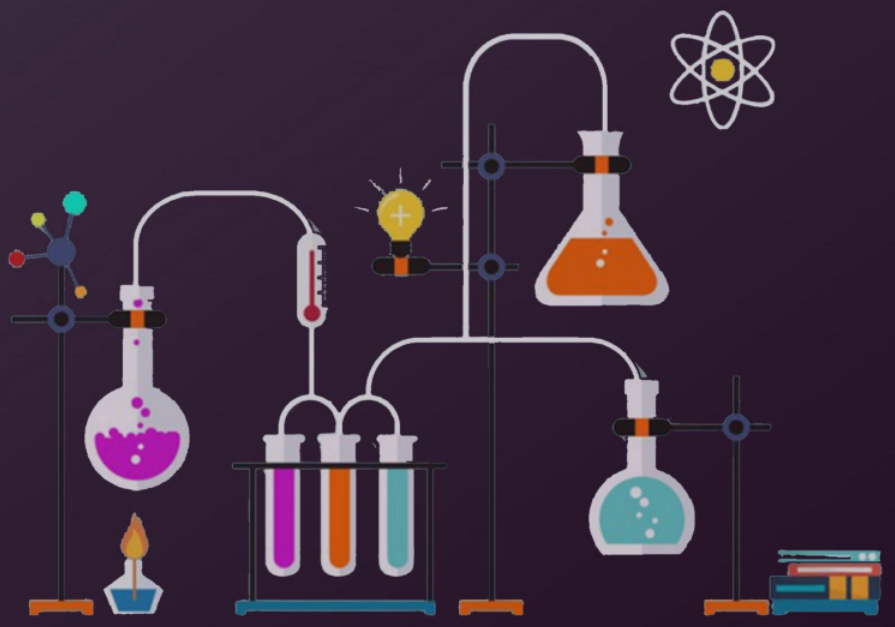
CHEMISTRY

Biomolecules, Polymers & Chemistry In Everyday Life



SHEET

eSaral.com

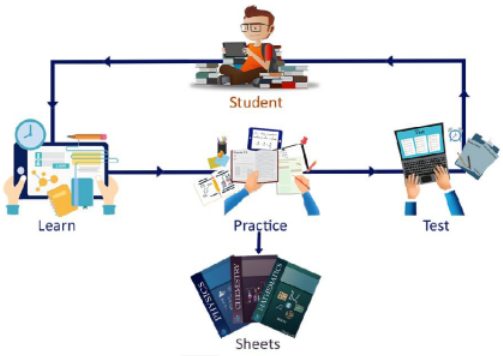
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 4
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 8
7.	Answer Key	Page - 13

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Classification of Carbohydrates and Epimers	1-2, 16-17, 26-28	2	12	
2	Glucose	3		1-3	
3	Fructose, Sucrose, Maltose and Lactose	18-19	4-5	4	1, 7, 18-19
4	Starch, Glycogen and Cellulose	4-5, 20-21, 29-30	11	13, 17-19	5-6, 8
5	Classification of Amino Acids	6-7	12-13	5-7, 14, 20	9-10
6	Zwitter Ion and Isoelectric Point of Amino Acids			8, 21	
7	Chemical Properties of Amino Acids	8-9, 22-25		22-23	20
8	Peptides	10		9	
9	Proteins	11-13, 31-34	6-10	15-16, 24	11-17
10	Nucleic Acids and Vitamines	14-15			
11	ACC		1, 3	10-11	2-4

Exercise - 1 *Concept Builder*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Classification of Carbohydrates and Epimers

Q1. Glucose and fructose are :-
(A) Chain isomers
(B) Position isomers
(C) Functional isomers
(D) Optical isomers

Q2. Oligosaccharides contain ……. Simple sugar units
(A) 2 to 10 (B) 4 to 8
(C) 6 to 12 (D) 6 to 10

Glucose

Q3. Glucose gives positive test with :-
(A) Tollen reagent (B) Fehling solution
(C) Bendict solution (D) All of these

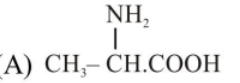
Starch, Glycogen and Cellulose

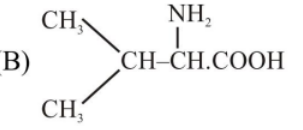
Q4. Cellulose is a polymer of :-
(A) Glucose (B) Fructose
(C) Ribose (D) Sucrose

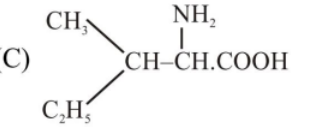
Q5. Cellulose is a linear polymer of
(A) α-glucose (B) β-glucose
(C) α-fructose (D) none of these

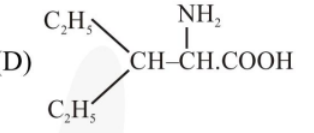
Classification of Amino Acids

Q6. The structural formula of an amino acid, isoleucine is :-

(A) 

(B) 

(C) 

(D) 

Q7. The simplest amino acid is
(A) glycine
(B) alanine
(C) guanine
(D) all of the above

Chemical Properties of Amino Acids

Q8. Peptide bond is :-
(A) $-COONH_4$ (B) $-N=C=O$
(C) $-CONH-$ (D) $-CONH_2$

Q9. At the isoelectric point, amino acid are present as
(A) H_2NCH_2COOH
(B) $H_3N^+CH_2COO^-$
(C) $H_3N^+CH_2COOH$
(D) $H_2NCH_2COO^-$

Peptides

Q10. Peptide bond is
(A) – CO – NH –
(B) NH_2 – CO – NH – R
(C) R – CO – NH – R
(D) – $CONH_2$

Proteins

Q11. α- Helix refers to :-
(A) Primary structure of Protein
(B) Secondary structure of Protein
(C) Tertiary structure of Protein
(D) Quanternary Structure of Protein

Q12. β-pleated strcuture of proteins is :-
(A) Secondary structure
(B) Primary structure
(C) Tertiary structure
(D) Quaternary structure

Q13. The helical structure of protein is established by :-
(A) peptide bonds
(B) Dipeptide bond
(C) Hydrogen bonds
(D) Ether bonds

Nucleic Acids and Vitamines

Q14. Which of the following bases is not present in a RNA?
(A) Uracil
(B) Thymine
(C) Adenine
(D) Guanine

Q15. Which of the following vitamins is not soluble in water ?
(A) C
(B) B_1
(C) B_2
(D) D

POLYMERS

Classification of Polymers

Q16. Teflon, polystyrene and neoprene are all
(A) Copolymers
(B) Condensation polymers
(C) Homopolymers
(D) Monomers

Q17. The polymer having strongest intermolecular forces is:
(A) Fibres
(B) Elastomer
(C) Thermoplastic
(D) Thermosetting polymer

Free Radical Mechanism of Addition Polymerisation

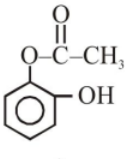
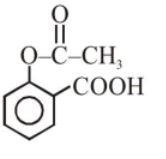
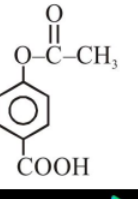
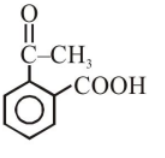
Q18. Example of addition co-polymer is :-
(A) Buna-S (B) Neoprene
(C) Nylon-66 (D) Dacron

Q19. Which of the following is not an addition polymer:-
(A) Bakelite (B) P.A.N.
(C) P.V.C. (D) Teflon

Rubber

Q20. The widely used plastic PVC is a polymerization product of
(A) $CH_2=CH_2$ (B) $CH_2=CHCl$
(C) $CHCl=CHCl$ (D) $CH_2=CCl_2$

Analgesics

Q21. Aspirin is :-
(A) 
(B) 
(C) 
(D) 

Resins and Polyesters

Q22. Which is condensation polymer :-
(A) Polythene (B) Styrine
(C) Bakelite (D) All of these

Q23. Bakelite polymer is formed by the polymerisation of methanal with :
(A) Salisaldehyde
(B) Hydroxy benzene
(C) Cinnamaldehye
(D) Azobenzene

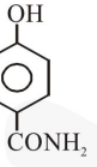
Q24. Terylene (Dacron) is a condensation polymer of
(A) Formaldehyde and urea
(B) Ethylene glycol and ethylene diisocyante
(C) Ethylene glycol and terephthalic acid
(D) Maleic anhydride and methyelene glycol

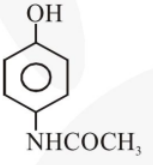
Q25. Polystyrene is a :
(A) Copolymer
(B) Addition polymer
(C) Condensation polymer
(D) None

CHEMISTRY IN EVERYDAY LIFE

Classification of Drug and Drug Target Interactions

Q26. The correct structure of drug paracetamol is –

(A) 

(B) 

(C) 

(D) 

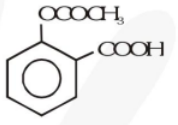
Q27. Which of the following types of drugs reduces fever :-
(A) Antipyertic (B) Analgesic
(C) Tranquiliser (D) Antibiotic

Q28. An antipyretic is –
(A) Quinine (B) Paracetamol
(C) Luminal (D) Piperazine

Analgesics

Q29. 2-Acetoxybenzoic acid is called –
(A) Antiseptic (B) Aspirin
(C) Antibiotic (D) Mordant dye

Q30. Compound A given below is :



(A) Antiseptic (B) Antibiotic
(C) Analgesic (D) Pesticide

ACC

Q31. Which of the following compounds is aspirin –
(A) Methyl salicylate
(B) Acetylsalicylic acid
(C) Phenyl salicylate
(D) Salicylic acid

Q32. Which of the following is an antidiabetic drug –
(A) Insulin (B) Penicillin
(C) Chloroquine (D) Aspirin

Q33. Arsenic drugs are mainly used in the treatment of–
(A) Jaundice (B) Typhoid
(C) Syphilis (D) Cholera

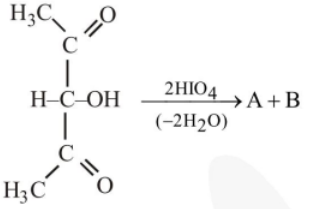
Q34. The drug given during hypertension is –
(A) Streptomycin (B) Chloroxylenol
(C) Equanil (D) Aspirin

Exercise - 1A — Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1.



A and B are

(A) A is CH_3COOH

(B) B is HCHO

(C) A is CH_3COCH_3

(D) B is HCOOH

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q2. **Statement-1:** Carbohydrates are defined as polyhydroxy aldehydes or Ketones or substances which give these on hydrolysis.
Statement-2 : Sucrose is not a carbohydrate.

SUBJECTIVE TYPE

Q3. Prove by giving one chemical reaction that glucose has hydroxyl (–OH) groups.

POLYMERS

MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q4. Which of the following are not copolymer
(A) Nylon-6 (B) Nylon-6,6
(C) Bakelite (D) Polyethene

Q5. Which of the following is/are addition polymer/s ?
(A) PVC (B) Nylon-6
(C) Teflon (D) Terylene

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 6 to 8

Polymers are high molecular weight compounds, whose structures are composed of large number of simple repeating units. The repeating units are usually low molecular weight compounds known as monomers. One of the methods of classifying polymers is into natural and synthetic polymers.

Q6. Which of the following is a natural polymer –
(A) Isoprene (B) Neoprene
(C) Chloroprene (D) Rubber

Q7. Which is a natural protein polymer –
(A) Polyisoprene (B) Keratin
(C) Chlorophyll (D) Cellulose

Q8. Which is not a natural polymer –
(A) Polyisoprene (B) Polysaccharide
(C) Collagen (D) Buna-S

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. Match the column –

Column I	**Column II**
(A) Melanine formaldehyde	(P) Non-breakable plastic cookery
(B) Glyptal	(Q) Nonstick utensil coating
(C) Polytetrafluoro ethylene	(R) Water proof shoes
(C) Buna-S	(S) Paints and lacquers

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q10. **Statement 1 :** Natural rubber is an addition polymer.
Statement 2 : Natural rubber has cis structure.

SUBJECTIVE TYPE

Q11. Arrange the following polymers in increasing order of their intermolecular forces. Also classify them as addition and condensation polymers : Nylon-66, Buna-S, Polyethene

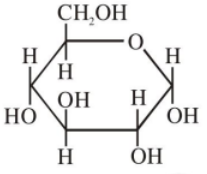
Q12. Examine the structure of the following synthetic polymers. Tell what class of compound each belong to and give structure of most likely monomers.
(a) Nylon-66 (fibres)

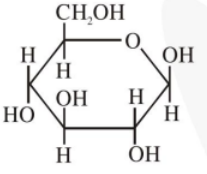
Exercise - 2 — Brain Booster

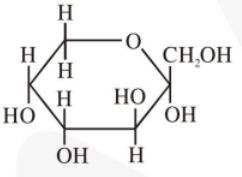
SINGLE CORRECT TYPE

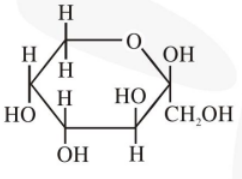
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Which of the following structures represents α-D-glucose ?

(A) 

(B) 

(C) 

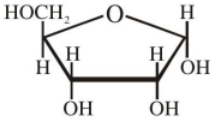
(D) 

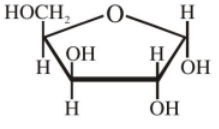
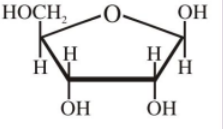
Q2. Which is not a reducing sugar :-

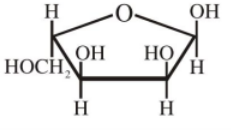
(A) Fructose (B) Glucose

(C) Lactose (D) Sucrose

Q3. Which of the following represents the anomer of the compound shown ?



(A)  (B) 

(C)  (D) None of these

Q4. The open-chain fructose on oxidation with HIO_4 gives-

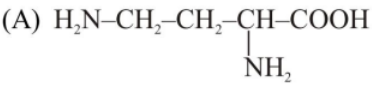
(A) $3\ HCOOH + 2\ H_2C=O + CO_2$

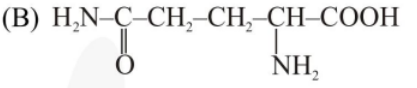
(B) $5\ HCOOH + H_2C=O$

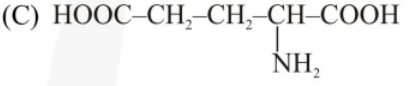
(C) $2\ HCOOH + 2\ H_2C=O + 2CO_2$

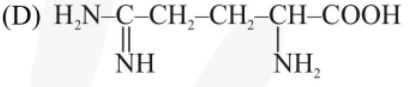
(D) $HCOOH + 3H_2C=O + 2CO_2$

Q5. Neutral amino acid is :

(A) 

(B) 

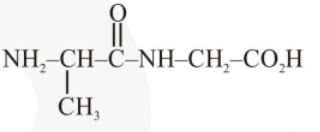
(C) 

(D) 

Q6. Which amino acid is not optically active ?

(A) Alanine (B) Glycine

(C) Phenyl alanine (D) Methionine

Q7. 

Identify the amino acid obtianed by hydrolysis of the above compound :

(A) Glycine

(B) Alanine

(C) Both (A) and (B)

(D) None of these

Q8. The structure of glycine (amino acid) is  (Zwitter ion).

Select the correct statement of the following :

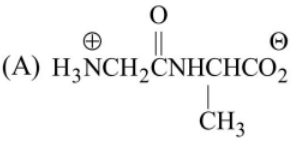
(A) Glycine, as well as other amino acids are amphoteric.

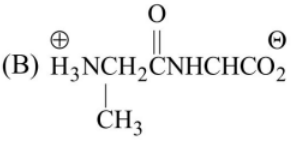
(B) The acidic functional group in amino acids is $-\overset{+}{N}H_3$

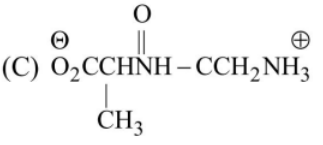
(C) The basic functional group in amino acids is $-CO_2^-$

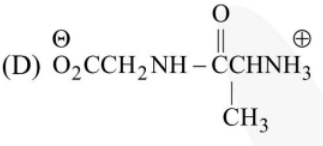
(D) All the statements are correct

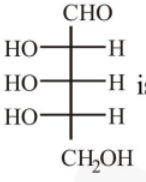
Q9. The dipeptide, Gly. Ala has structure –

(A) 

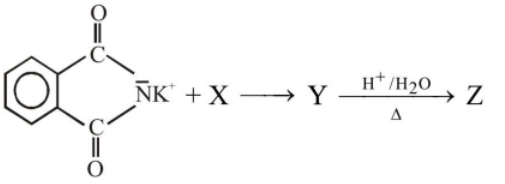
(B) 

(C) 

(D) 

Q10.  is

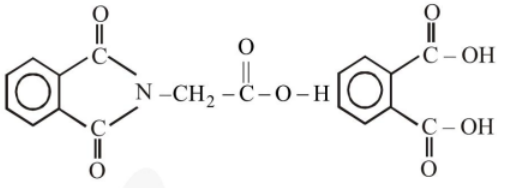
(A) L(+)–Ribose
(B) L–(+)–Arabinose
(C) L–(–) Xylose
(D) L–(+)–Lyxose

Q11. 

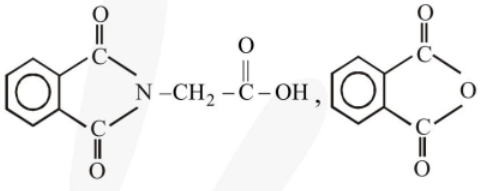
+ Glycine

Compound X, Y, Z are-

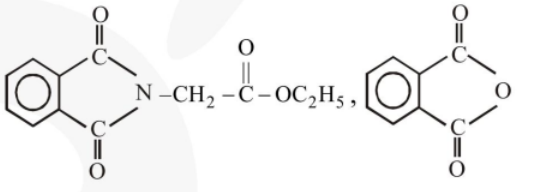
(A) Chloroacetic acid,



(B) Bromo acetic acid



(C) Ethyl chloroacetic acid,



(D) All of the above.

POLYMERS

Q12. Which of the following is a copolymer
(A) Natural rubber (B) Polystyrene
(C) PAN (D) Bakelite

Q13. Natural rubber is a polymer of :-
(A) Trans-isoprene
(B) Cis-isoprene
(C) Cis and trans isoprene
(D) None of these

Q14. The fibre obtained by the condensation of hexamethylene diamine and adipic acid is:
(A) Dacron (B) Nylon 6,6
(C) Rayon (D) Teflon

Q15. Orlon is a polymer of :-
(A) Tetra fluoroethene
(B) Acrylonitrile
(C) Acetic acid
(D) Benzene

Q16. An example of a condensation polymer is
(A) PVC
(B) terylene
(C) polypropylene
(D) polystyrene

CHEMISTRY IN EVERYDAY LIFE

Q17. Paracetamol is an –
(A) Both antipyretic and analgesic
(B) Analgesic
(C) Antipyretic
(D) Antimalarial

Q18. Aspirin is called –
(A) Pyretic (B) Antiseptic
(C) Antibiotic (D) Antipyretic

Q19. Aspirin is an acetylation product of –
(A) p-dihyroxybenzene
(B) o-hydroxybenzoic acid
(C) o-dihydroxy benzene
(D) m-hydroxybenzoic acid

Q20. Chloramphenicol is an –
(A) Analgesic (B) Anaesthetic
(C) Antibiotic (D) Antiseptic

Q21. Phenol is :-
(A) Antimicrotrial (B) antiseptic
(C) disinfectant (D) All of these

Q22. Sulpha drugs are derivatives of –
(A) Benzene sulphonic acid
(B) Sulphanillic acid
(C) Sulphanilamide
(D) p - aminobenzoic acid

Q23. Detergents are prepared by the action of H_2SO_4 followed by neutralization by starting with -
(A) Cholesterol (B) Lauryl alcohol
(C) Cyclohexanol (D) p-Nitrophenol

Q24. Which one of the following types of drugs reduces fever-
(A) Tranquilizer (B) Antibiotic
(C) Antipyretic (D) Analgesic

Exercise - 2A *Brain Booster*

ONE ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is/are not disaccharide?
(A) Sucrose (B) Glucose
(C) Fructose (D) Starch

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Parafraph for Q. 2 to 4

Carbohydrates are polyhydroxy aldehydes and ketones and those compounds which on hydrolysis give such compounds are also carbohydrates. The carabohydrates which are not hydrolysed are called monosaccharides. Other carbohydrates are oligosaccharides and polysaccharides. Monosaccharides with aldehydic group are called aldose and those with free ketonic groups are called ketose. Carbohydrates are optically active.

Number optical isomers = 2"

where n = number of asymmetric carbons. Carbohydrates are mainly synthesised by plants during photosynthesis.

Q2. Number of optical isomers of glucose is :
(A) 8 (B) 16
(C) 32 (D) 4

Q3. Fructose is :
(A) aldotriose (B) ketohexose
(C) aldohexose (D) Ketotriose

Q4. First member of ketose sugar is :
(A) ketotriose (B) ketotetrose
(C) ketopentose (D) ketohexose

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q5. Match the column –

Column I	**Column II**
(A) α-D glucose and β-D glucose	(P) Enantiomer
(B) D (+) glucose and L (–) glucose	(Q) Diastereomer
(C) D (+) glucose and L (–) galactose	(R) Epimers
	(S) Anomers

Q6. Match the column I with column II.

Column-I	**Column-II**
(A) Sucrose	**(P)** Non-reducing sugar
(B) Fructose	**(Q)** Polysaccharide
(C) Maltose	**(R)** Reducing sugar
(D) Cellulose	**(S)** Disaccharide

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q7. **Statement 1 :** Maltose is a reducing sugar while sucrose is not.
Statement 2 : Maltose is a monosaccharide and sucrose is a disaccharide.

POLYMERS

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q8. Incorrect statements related to vulcanization of rubber
(A) Sulphur reacts to form a new compound
(B) Sulphur cross-links are introduced
(C) Sulphur forms a very thin protective layer over rubber
(D) All statements are correct

Q9. Condensation of caprolactum does not gives
(A) Nylon-6,6
(B) Nylon-6
(C) Nitrile rubber
(D) Nylon-6,10

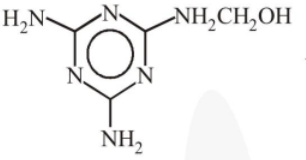
Q10. Nylon material is preferred over materials of cotton or silk because –
(A) dirt can be washed off easily
(B) can be dyed with different colours
(C) of durability
(D) they are not affected away by insects

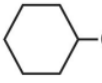
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

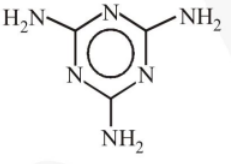
Paragraph for Q. 11 to 14

Observe the following reactions :

(i) A + HCHO $\longrightarrow$  $\xrightarrow{\text{Polymerisation}}$ B

(ii)  $\xrightarrow{[O]}$ [C] $\xrightarrow{NH_2OH}$ [D] $\xrightarrow{H_2SO_4}$ [E] $\xrightarrow{H_3O^{\oplus}}$ [F] $\xrightarrow{\Delta}$ [G]

Q11. Structure of (A) is –

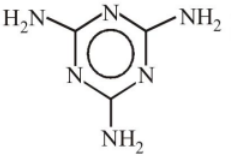
(A) 

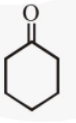
(B) 

(C) $H_2N-(CH_2)_5-COOH$

(D) $\left(-NH-(CH_2)_5-\overset{\overset{O}{\|}}{C}-\right)_n$

Q12. Structure of (C) is –

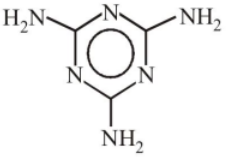
(A) 

(B) 

(C) $H_2N-(CH_2)_5-COOH$

(D) $\left(-NH-(CH_2)_5-\overset{\overset{O}{\|}}{C}-\right)_n$

Q13. Structure of (F) is –

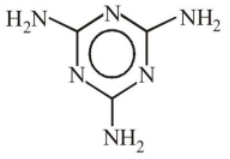
(A) 

(B) 

(C) $H_2N-(CH_2)_5-COOH$

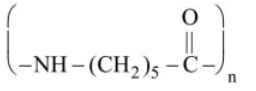
(D) $\left(-NH-(CH_2)_5-\overset{\overset{O}{\|}}{C}-\right)_n$

Q14. Structure of (G) is –

(A) 

(B) 

(C) $H_2N-(CH_2)_5-COOH$

(D) 

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST I. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q15. Match the items of list I with those of list II. Select the correct answer using the coeles given below the lists :-

	List-I (Polymers)		List-II (Monomers)
(A)	BUNA-n	(P)	Phthalic acid and ethylene glycol
(B)	Nylon-66	(Q)	Terephthalic acid and ethylene glycol
(C)	Dacron	(R)	Hexa methylenediamine and adipic acid
(D)	Glyptal	(S)	Acrylo nitrile and 1, 3-butadiene

Codes :-

(A) A-S B-R C-P D-Q

(B) A-S B-R C-Q D-P

(C) A-R B-S C-Q D-P

(D) A-Q B-R C-S D-P

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q16. Match the column –

Column I	**Column II**
(A) Neoprene	(P) Synthetic
(B) Bakellite	(Q) Addition
(C) Orlon	(R) Condensation
(D) Nylon-66	(S) Copolymer

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

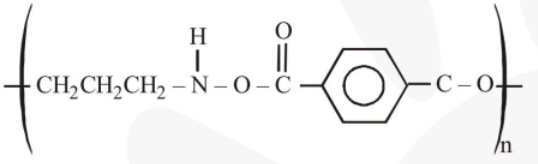
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q17. **Statement 1 :** Nylon 6,6 is a fibre.
Statement 2 : It is an addition polymer.

SUBJECTIVE TYPE

Q18. Poly(butylene terephthalate) is hydrophobic plastic material widely used in automotive ignition systems.



(a) What type of polymer is poly (butylene terephthalate) ?

(b) Is this an addition polymer or a condensation polymer ?

(c) Suggest, what monomers might be used to synthesize this polymer and how the polymerization might be accomplished.

Q19. Name the monomers with structures for the following polymers :
(i) Nylon-66
(ii) Neoprene

Q20. Write the information asked for in the following polymers :
(i) Bakelite – Materials used for preparation.
(ii) PVC – Monomer units.
(iii) Nylon-66 – Materials required for preparation.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	A	D	A	B	C	A	C	B	A
11	12	13	14	15	16	17	18	19	20
B	A	C	B	D	C	A	A	A	B
21	22	23	24	25	26	27	28	29	30
B	C	B	C	B	B	A	B	B	C
31	32	33	34						
B	A	C	C						

Ex-1A

1. A, D **2.** C **4.** A, D **5.** A, C **6.** D
7. B **8.** D
9. A → P; B → S; C → Q; D → R
10. B

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	B	A	B	B	D	D	A	A
11	12	13	14	15	16	17	18	19	20
C	D	B	B	B	B	A	D	D	C
21	22	23	24						
D	C	B	C						

Ex-2A

1. B, C, D **2.** B **3.** B **4.** A
5. A → S, Q; B → P; C → R, Q
6. (A) → P, S ; (B) → R ; (C) → R, S ; (D) → P, Q
7. C **8.** A, C, D **9.** A, C, D **10.** A, B, C, D **11.** A
12. B **13.** C **14.** D **15.** B
16. A → P,Q; B → P, R, S; C → P, Q; D → P, R, S **17.** C

-: Important Notes & Formulas :-